

CHAPTER 4

Introduction to Refining

How a refinery turns one stream of crude into finished fuels — through three core processes: distillation, conversion and treatment.



WHY REFINE AT ALL?

Raw crude is almost useless

Straight from the ground, crude oil cannot run a car, fly a plane or heat a home. It is a thick mixture of thousands of different hydrocarbon molecules, all jumbled together — valuable only for what can be separated out of it.

A refinery is the industrial plant that unlocks that value. Running continuously, around the clock, it splits, rebuilds and cleans the barrel into the precise fuels and feedstocks the world actually buys.

200k – 1M+

barrels a day in a single large refinery — processed non-stop

A continuous process

A refinery never really stops — shutting down and restarting is slow and costly, so units run 24/7 for years between turnarounds.

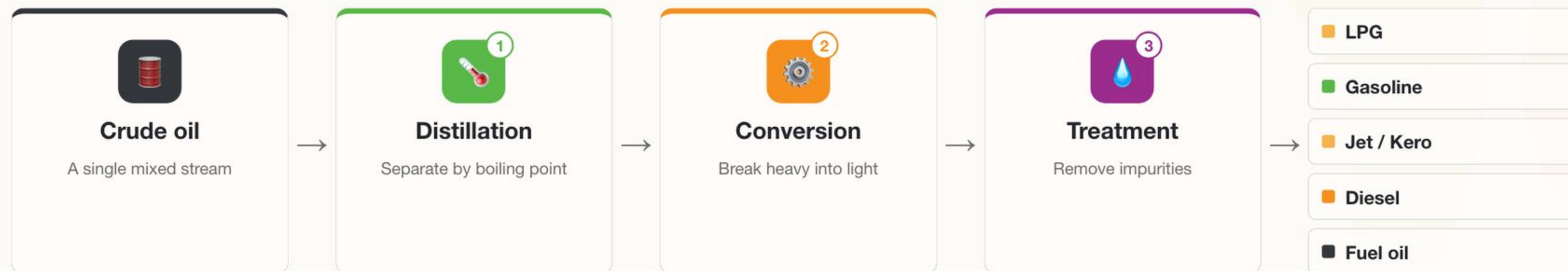
Three jobs, in order

Separate the barrel (distillation), rebuild the heavy part (conversion), then clean every stream (treatment) before it is sold.

THE REFINERY AT A GLANCE

One crude in, a slate of products out

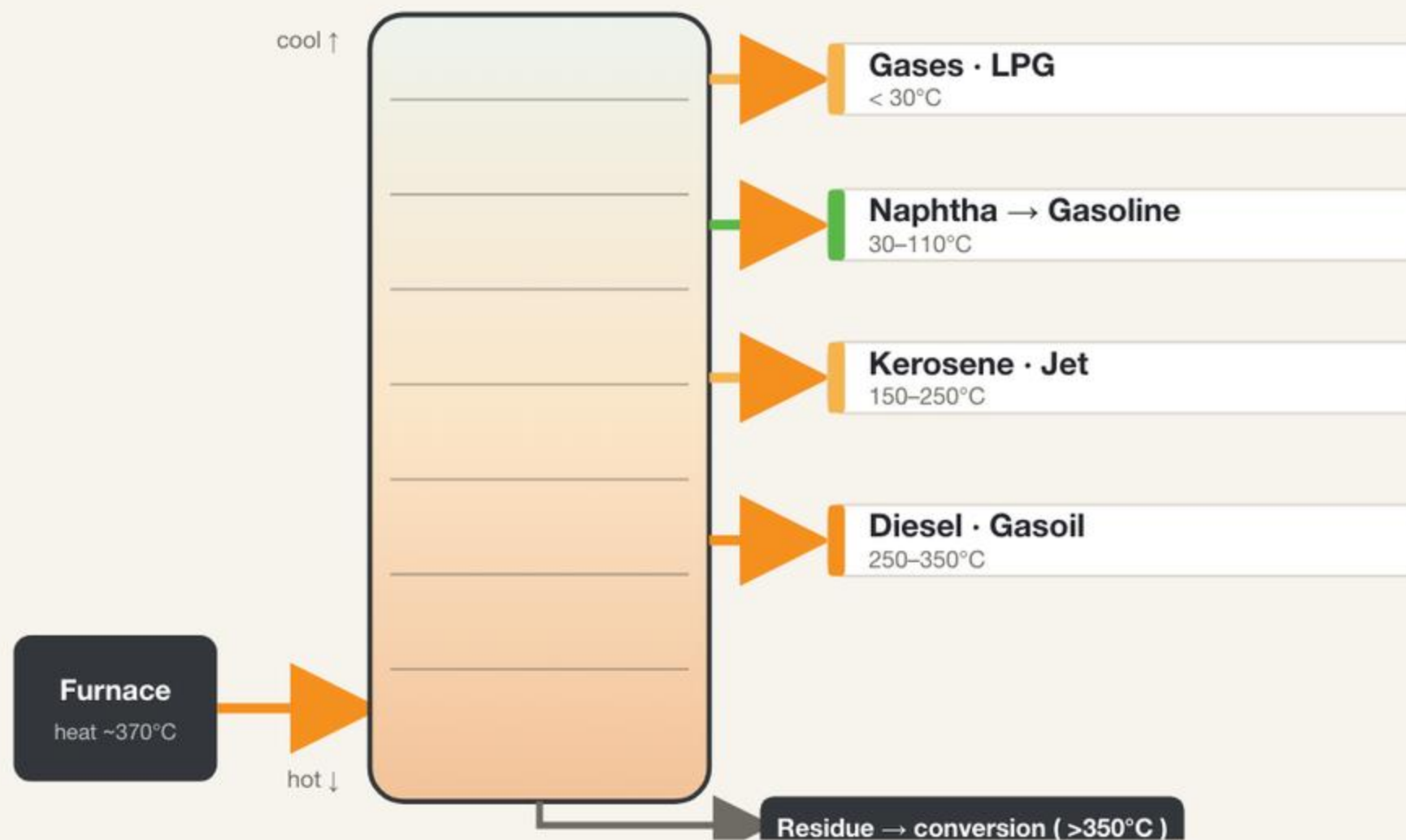
However large or sophisticated, every refinery is built from the same three steps in the same order. Crude enters as a single mixed stream and leaves as a dozen distinct, sellable products.



Distillation does the sorting, conversion does the upgrading, and treatment does the cleaning. **The rest of this chapter opens each step in turn.**

Distillation — separating by boiling point

The first and largest step. Crude is heated until it boils, then fed into a tall column where each fraction condenses at its own height — a purely physical sort, the backbone every refinery is built around.



Heat, then separate

Crude is heated in a furnace to around 350–370 °C until most of it turns to vapour, then enters the column near the base.

Light on top, heavy below

As vapours rise they cool. The lightest gases leave the top; each heavier fraction condenses lower down and is drawn off.

Nothing changes chemically

Distillation only sorts molecules already present. To make more of the valuable light ones, the barrel must be converted.

The distillation fractions

Each fraction leaves the column at a different height and goes on to a different market — from the lightest gases at the top to the heavy residue at the bottom.

FRACTION	BOILING RANGE	WHAT IT BECOMES
■ Refinery gas · LPG	< 30 °C	Fuel gas, propane and butane, petrochemical feedstock
■ Naphtha	30–110 °C	Gasoline blending and petrochemical feed (plastics)
■ Kerosene	150–250 °C	Jet fuel and heating kerosene
■ Gas oil	250–350 °C	Diesel and heating oil — the economy's fuel
■ Atmospheric residue	> 350 °C	Feed to the vacuum unit, or heavy fuel oil

TWO COLUMNS

Atmospheric and vacuum distillation

Distillation actually happens in two stages — the second one rescuing valuable product that the first has to leave behind.

Atmospheric distillation

The first and main column, run at normal pressure. It takes the bulk of the light gases, naphtha, kerosene and gas oil out of the crude in a single pass.

Vacuum distillation

The leftover residue goes to a second column held under vacuum. Lower pressure lowers boiling points, so more heavy gas oil can be recovered without overheating the oil.

Why pull a vacuum?

At normal pressure the residue would need to be heated so hot it would crack uncontrollably and foul the column. A vacuum sidesteps that, recovering extra value cleanly and safely.

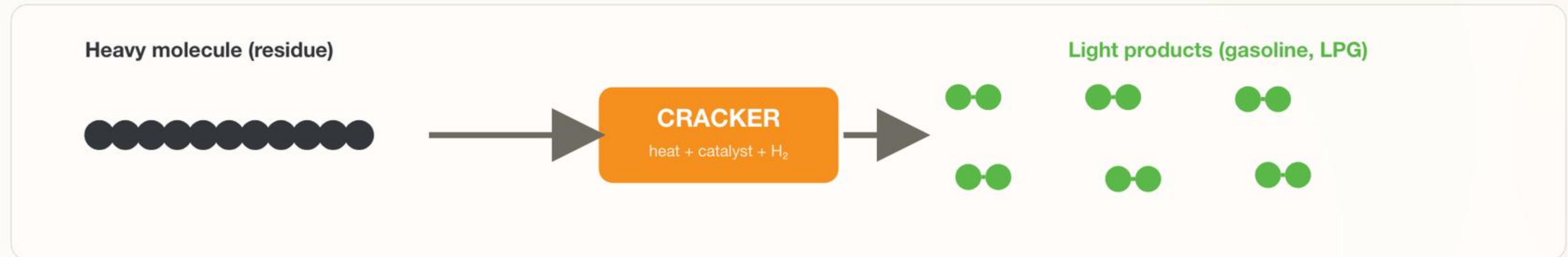
What is left at the bottom

The base of the vacuum column — vacuum residue — is the heaviest, dirtiest cut of all: feed for cokers, or the basis of bitumen and heavy fuel oil.

PROCESS 2 · ADDING VALUE

Conversion — breaking heavy into light

Distillation alone leaves a refinery with more heavy, low-value material than the market wants. Conversion fixes the mismatch: it chemically breaks big, heavy molecules into the small, light ones that make petrol and diesel.



The value step

Cracking splits large, cheap molecules into the smaller ones that make road fuel — directly lifting the dollar value of every barrel processed.

Why it matters commercially

Conversion is what lets a refiner buy discounted heavy crude and still turn out a high-value light product slate — the heart of refining economics.

THE TOOLKIT

The conversion units

Different units crack in different ways. A refinery's particular mix of them defines what it can make — and how heavy and sour a crude it can profitably run.



Catalytic cracker (FCC)

Uses a hot powdered catalyst to crack heavy gas oil into gasoline and LPG. The workhorse of most refineries, especially gasoline-focused ones.



Hydrocracker

Cracks under high-pressure hydrogen to produce very clean, high-quality diesel and jet. Flexible and powerful, but capital-intensive.



Coker

Attacks the very heaviest vacuum residue, breaking it into lighter products and leaving solid petroleum coke. The deepest conversion of all.

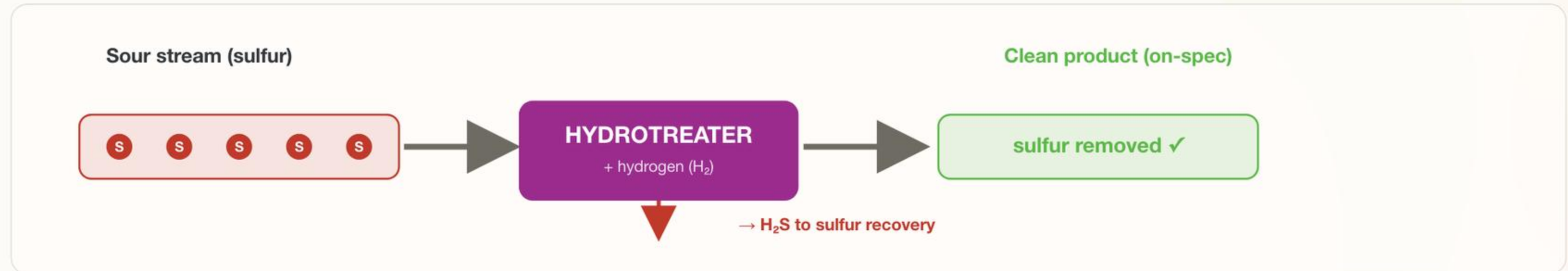


Reformer & alkylation

Rearrange light molecules to raise octane, building the high-quality blendstocks that finished gasoline needs to perform.

Treatment — removing the impurities

Even after distilling and converting, the streams still carry sulfur, nitrogen and metals. Treatment cleans them so each one can legally and safely be sold as a finished fuel.



From crude cut to finished fuel

Hydrotreating adds hydrogen to strip out sulfur, nitrogen and metals — turning a raw distillation cut into a stream that meets fuel law.

Driven by regulation

Ever-tighter sulfur limits on diesel, marine fuel and gasoline make treatment the step that keeps a refinery's products legal to sell.

THE FINAL ASSEMBLY

Blending — building the finished grade

A refinery rarely sells a single stream as-is. The last step blends several treated streams — plus additives — to hit the exact specification of a finished, market-ready grade.



Specs, not streams

Buyers don't want "FCC gasoline" — they want a grade that meets octane, sulfur, vapour-pressure and cold-flow limits exactly. Blending hits those numbers.

Regional & seasonal recipes

The same refinery blends different recipes for summer vs winter, and for each market's rules — which is why fuels are not interchangeable everywhere.

NOT ALL REFINERIES ARE EQUAL

Simple vs complex refineries

How much conversion a refinery has decides which crude it can run and how much value it can wring out of each barrel.

LOW COMPLEXITY

NCI ~1–4

Hydroskimming

Distillation and treating, with little or no conversion. It needs light-sweet crude and yields a lot of low-value fuel oil — thin, exposed margins.

NELSON
COMPLEXITY



HIGH COMPLEXITY

NCI ~10–12+

Deep conversion

Adds crackers and cokers on top. It can run cheap heavy-sour crude and still squeeze out a high-value light slate — wider, more defensible margins.

The Nelson Complexity Index scores how much upgrading a plant can do. Higher complexity means more freedom on feedstock — and a structurally better margin.

FROM PLANT TO MARKET

Why refining matters to a trader

You may never set foot in a refinery — but its three processes shape the prices you trade every single day.

It creates the products

Refining is the bridge between crude and the fuels the world buys — the reason a crude market and a product market exist side by side.

It sets the crack spread

The value these three processes add is exactly what the crack spread measures — and what refiners hedge in the financial market.

It prices crude quality

Because complex refineries can use heavy-sour crude, conversion capacity is what gives those discounted grades any value at all.

It drives product spreads

Outages, run cuts and maintenance turnarounds in these units move product prices — among the events traders watch most closely.

CONCLUSION — THREE STEPS, ONE BARREL

Refining is where crude becomes value.

Distillation separates

It sorts the barrel into fractions by boiling point — but changes nothing chemically.

Conversion creates

Cracking and coking break heavy molecules into the light products that carry the value.

Treatment & blending finish

Cleaning and blending turn raw cuts into the exact, on-spec fuels the market will buy.